Write a program to implement Shortest Job First (SJF) Preemptive Scheduling for three processes and calculate the total context switches and average waiting time. The processes have burst times 10ns, 20ns, and 30ns,arriving at 0ns, 2ns, and 6ns, respectively

main.c

Run

```c
#include <stdio.h>
#include <limits.h>

int main() {
    int n = 3;
    int bt[3] = {10, 20, 30};     // Burst times
    int at[3] = {0, 2, 6};        // Arrival times
    int rt[3];                    // Remaining times
    int wt[3] = {0};              // Waiting times
    int ct[3];                    // Completion times

    int time = 0, completed = 0;
    int min_rt, shortest = -1;
    int prev = -1; // to count context switches
    int context_switches = 0;

    // Copy burst time to remaining time
    for (int i = 0; i < n; i++) {
```

Output

Clear

```
Process Arrival Burst   Waiting Completion
P1  0   10  0   10
P2  2   20  8   30
P3  6   30  24  60

Total Context Switches = 2
Average Waiting Time = 10.67 ns


=== Code Execution Successful ===
```